

Default Risk and Borrowing Limits Across Firms and Over Time

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ABSTRACT

We introduce earnings based borrowing limits into a business cycle model where firms finance investment using debt subject to default risk. In contrast to existing models studying aggregate fluctuations in environments where firms may default on loans, our work is consistent with both the empirical size over age distribution of firms and the survival rates of young firms. We show that matching the size over age distribution without borrowing constraints would imply failure rates far higher than seen in the data. The size over age distribution has an important role in determining the aggregate costs of financial frictions, both in the loan run and over the business cycle.

Loan rates that rise with default risk, and therefore leverage, alongside borrowing limits that are particularly severe for productive but small firms, shift investment towards larger, older firms, with higher earnings and greater collateral. There is a misallocation of capital across firms and young, productive firms operate at lower levels than older counterparts with the same productivity. As young firms tend to be small and heavily reliant on external finance, financial frictions discourage entry, reducing the number of production locations. This leads to a further increases in misallocation. Measured aggregate total factor productivity falls by 15 percent, on average, when compared to an economy without costly external finance.

A financial shock drives a recession similar to the U.S. 2007 recession in several respects. A combination of higher costs of borrowing, alongside a deterioration of firms' balance sheets, increases the misallocation of production causing a sharp downturn and protracted recession. There is a disproportionate loss of employment in young, productive firms and exit rates rise while entry falls. There is a disparate effect on cash-poor firms increasing in misallocation. The negative impact on firm values boosts default and exit while suppressing entry, driving large, persistent losses in the number of firms. Measured TFP falls over several periods, as do investment and GDP, and the ultimate declines in investment and employment are large relative to that in TFP. The recovery that follows is gradual given slow repair to aggregate capital, its distribution across firms and the stock of firms. We show that the real aggregate effects of a financial shock depends critically on the size-age distribution of firms.

1 Introduction

Several empirical anomalies characterise the Great Recession recession when compared to the typical post-war U.S. recession. There was a large fall in GDP relative alongside a moderate fall in measured total factor productivity. There was an atypical fall in investment relative to the decline in output. The economy experienced a sharp contraction in aggregate lending uncommon across previous recessions. Beyond macroeconomic aggregates, disaggregated data reveal an unusual disparity in the impact on firms. Large declines in firm entry rates alongside losses disproportionately concentrated in small and young firms that are more reliant on external finance suggest that importance of a shock affecting firms' ability to borrow.¹ At the same time, the corporate sector as a whole held unusually large levels of cash at the start of the recession.² These observations suggest that a quantitative analysis of the real effect of a financial shock should include a nontrivial distribution of firms that vary in their assets and debt as well as productivity.

Building a dynamic stochastic general equilibrium model with a time-varying distribution of firms, we study the role of both aggregate shocks to the exogenous component of total factor productivity and shocks to the costs of external finance. Our model is consistent with macroeconomic aggregates, the size-over-age distribution of firms and the aggregate debt-to-asset ratio. The model economy exhibits considerable heterogeneity in the availability and costs of loan across firms. These costs vary endogenously with firm age and size, and aggregate shocks that affect the costs of credit have disparate incidence across the distribution of firms. The result is a substantial rise in the misallocation of capital not observed following an exogenous shock to total factor productivity.

Credit shocks, in our model, lead to a deterioration in firms' cash positions and raise the costs of borrowing. We find that such shocks are able to explain several of the empirical anomalies seen in the Great Recession. A credit shock consistent with the observed reduction in lending generates a small fall in total factor productivity alongside a large drop in GDP and a unusually severe fall in investment. These declines are considerably larger than a comparable shock to total factor productivity. The latter fails the fall of investment relative to output, and generates little decline

¹Using quarterly BED data maintained by the Bureau of Labor Statistics, Khan and Thomas (2013) find firms with fewer than 100 employees, as a whole, contracted employment twice as much as large firms and accounted for 46 percent of total employment losses, far beyond their 38 percent average employment share. The average relative size of a firm aged 0 through 5 years in the Business Dynamics Statistics database fell 4.2 percentage points over 2007 - 2015. Elsewhere, Almeida et al. (2009) find that firms that had to refinance a significant fraction of their debt in the year following August 2007 experienced a one-third fall in their investment relative to otherwise similar firms.

²Bates et al (2009) show cash held among nonfinancial firms rose through 2005; see also Khan and Thomas (2013).

in lending. Most importantly, in our environment, a credit shocks delivers the disproportionate contractions among small firms, young firms and firms more reliant on external finance, as well as the unusual reduction in the number of operating firms. A TFP shock generates little rise in exit and no difference in the decline in economic activity across small and large firms.

Our results hinge on reproducing the size-over-age distribution of firms, a result of introducing earnings based borrowing limits to young firms that disappear over time. Thus the model has both defaultible debt and, beyond it limits on the levels of debt certain firms may consider. Our environment is unique in featuring a distribution of firms over capital, debt and persistent idiosyncratic productivity, endogenous entry and exit, and shocks influencing default risk and the loan rate schedules offered to borrowers. It is further distinguished by qualitative differences in the business cycles it predicts following a financial shock versus a real shock. It is yet more rare in the sensitivity of its financially-led recession and recovery episodes to changes in the average underlying age-size distribution of firms, with no such sensitivity during productivity-shock driven episodes.

In our model, firms may default on non-contingent debt. When such default occurs, creditors recover only a fraction of the firm's collateral, its capital. Competitive lenders choose loan rate schedules for each loan delivering them an expected return equal to the real return on risk-free debt, so the interest rate charged on any given loan rises with the probability of default and falls in the collateral recoverable under default. Firms' probabilities of default vary with their expected future productivity, capital stock and debt; thus, so do their costs of borrowing. Furthermore, firms ability to refinance debt, and thus the risk associated with their current borrowing, is affected by earnings based borrowing limits. All firms in the economy enter with such limits, they disappear randomly over time.

By assuming that a defaulting firm loses its assets and must immediately exit, we can derive several results allowing us to characterize borrowing and lending in general equilibrium. First, absent firm-level capital adjustment costs, we show that idiosyncratic productivity and cash-on-hand are sufficient to describe a firm's state. Second, we show that firm values are continuous and weakly increasing in cash. Third, we establish the existence of cash thresholds with which we can fully summarize firms' decision rules regarding default and thereby isolate equilibrium loan schedules. Specifically, given the current aggregate state, we derive threshold cash levels varying with idiosyncratic productivity, with each representing the minimum cash-on-hand such that a firm realizing a given productivity will repay its loan and continue operating. Importantly, in our setting, the timing of default implies that these threshold levels of cash must be jointly determined alongside

firm values. The implication is that loan rates schedules must be solved alongside firm values; they are not known a priori. We exploit these results to numerically characterize borrowing and lending in our model where the aggregate state includes the distribution of a time-varying number of firms over idiosyncratic productivity, capital and debt.

In general, firms with higher expected future earnings or capital have a lower risk of default and so can borrow more, and at lower cost. Nonetheless earnings based borrowing constraints may still limit such firms use of debt. Given any feasible level of debt, the risk of default rises with leverage. We characterize firm decision rules conditional on membership in one of three groups, and then identify threshold cash levels determining membership in these groups. Firms having stockpiled sufficient cash to ensure they can invest efficiently in every possible future date and state without ever again incurring default risk are henceforth *unconstrained* to financial shocks; these firms invest efficiently, borrow at the risk-free rate and follow debt-dividend policies ensuring they can always do so. Were it not for exogenous death risk, such firms would live forever and ultimately take over all production. Outside this group, firms with adequate cash to invest efficiently in the current period with no immediate default risk take on the necessary loans to do so at the risk-free interest rate; these firms pay no dividends in efforts to reach the absorbing unconstrained state. The remaining group of firms face risk-premia to invest efficiently, so their investments are affected by their cash; they also face immediate default risk and so may exit the economy, eliminating useful production locations.

We calibrate our model using firm-level employment and financial data, on default rates and debt recovery rates, real aggregate quantity data, data from the flow of funds, and evidence on the frequency and duration of financial crises. Turning to the distribution of firms over age seen in the data, we choose operating costs and borrowing limits to match the growth and survival rates of entrants and young firms. We show that borrowing limits and risky debt leads to a long-run misallocation of resources relative to a setting without financial frictions. Larger firms with more collateral invest more than smaller firms with less collateral on average, although they draw from the same productivity distribution. The implication of this is that insufficient capital is allocated to small firms. That in turn discourages entry, reducing the number of production locations, since young firms tend to be small and heavily reliant on external finance. This misallocation, alongside endogenous exit in the event of default, reduces average aggregate total factor productivity, alongside GDP and its inputs.

Examining business cycle moments from our model driven by both real and financial shocks, we

find that they resemble those from a version of the model without financial shocks, which are in turn similar to those from a frictionless representative agent model. The latter is because aggregate TFP shocks affect our firms in a largely even way, yielding little change in the extent of misallocation. The former is due to the infrequent nature of financial shocks, not because their consequences resemble a TFP-driven recession.³ When our economy experiences exogenous shocks to aggregate TFP, non-contingent loans drive countercyclical default and so are reinforced by modest endogenous TFP effects. Nonetheless, irrespective of our model’s micro-level calibration, its aggregate quantity responses following a shock to TFP closely resemble those from a representative agent model such as Hansen (1985). Thus, a productivity shock selected to yield the observed decline in measured TFP over the 2007 recession falls far short of explaining the observed declines in GDP, investment and employment. It also implies a reduction in lending an order of magnitude too low, and it does not yield disproportionate employment declines among small and young firms.

When we examine the response to a financial shock in our model, the greatest declines in output, employment and investment do not occur at the onset. This distinction relative to the response following a TFP shock stems from a resulting allocative disturbance that compounds through time. Unlike an across-the-board productivity shock, a shock affecting firms’ cash positions and borrowing terms has disparate implications throughout the distribution of firms, concentrating immediate losses and lasting damage among firms most reliant on credit for their growth and survival.

Following a credit shock that raises loan rates and damages firm balance sheets, the model economy goes through a large, slowly unfolding recession then a slow recovery. While we assume that credit conditions are restored in 4 periods, GDP remains substantially below trend 20 periods later. At the shock’s impact, a larger than usual set of young, cash-poor firms with relatively high productivities that should be growing rapidly find themselves undertaking suboptimal investments, with these investments further below optimal than in normal times. As a result, beginning with the first date thereafter, the allocation of capital, and thus production, across incumbent firms is unusually distorted. This problem worsens with time as new generations of firms are affected and each generation requires more time to reach efficient mature size. Furthermore, because the shock causes a sharp fall in the values of cash-poor firms, and entrants tend to be cash-poor, there are fewer new firms than usual entering production in each period while exit rates rise, particularly among young firms made increasingly vulnerable by each date’s inefficiently low investment and production. Given decreasing returns-to-scale production, the resulting destruction to the stock

³Large financial shocks are rare in the postwar U.S. (Reinhart and Rogoff (2009) and Bianchi and Mendoza (2012)).

of firms delivers a second form of capital misallocation in the form of lost production locations. Misallocation rises over time as these forces compound, so endogenous productivity and real activity gradually unravel, with the consequent reductions in the returns to saving driving large declines in investment, and hence employment, GDP and consumption. Thereafter, the recovery episode is far slower than what follows an aggregate TFP shock. Even by comparison to the Khan and Thomas (2013) setting where an aggregate shock tightens exogenous collateral constraints, our model economy exhibits significantly more gradualism in recovery due to its second, extensive-margin, channel of misallocation. When financial conditions revert to normal, it takes many periods to repair the effects of our two misallocation channels on the distribution and stock of firms. So long as the measure of producers has not recovered, the return on capital remains low, slowing the rise in aggregate investment and thus the recovery in aggregate capital, with that in turn hindering recovery in the stock of firms and its distribution.

It turns out that the microeconomic details of our model’s calibration that shape the usual firm age-size distribution can have first-order effects on its aggregate predictions following a financial shock.

2 Related work

Recent studies have focused on how costly external finance affects macroeconomic variables in heterogeneous firms operate subject to collateral constraints or borrow risky debt. Given decreasing returns to scale at the individual firm, a time-varying distribution of producers creates endogenous movements in aggregate total factor productivity. Khan, Sengupta and Thomas (2022) study the role of risky debt in an equilibrium business cycle model. They do not allow for earnings based borrowing limits and, as a result, generate far lower levels of misallocation. Using a similar model with a different timing of default and capital quality shocks, Ottonello and Winberry (2020) study the effect of monetary policy shocks. They find that firms without significant costs of borrowing are more responsive to monetary policy shocks. In their framework the measure of producers is constant over time as each defaulting firm is replaced by a new firm. Our emphasis is on aggregate effects of changes in the distribution of firms over the business cycle. In our model, the time-varying measure of producers amplifies the effect on aggregate measured TFP following a credit shock.

Our work is related to the seminal paper by Jermann and Quadrini (2012), which examined a representative firm model wherein investment is financed with debt and equity. Given limited enforceability of debt contracts, the firm faces endogenous limits on its debt issuance. That friction

is not trivially evaded using changes in equity, because there are convex costs associated with such adjustments. In this setting, financial shocks are shocks affecting the fraction of the firm’s value recoverable under default, and thus the severity of borrowing limits. Because the firm’s enforceability constraint always binds, a shock tightening this constraint causes large reductions in real economic activity. Beyond our emphasis on heterogeneity and equilibrium default, a notable difference in our setting is that financial frictions do not dampen the response of the aggregate economy to non-financial shocks.

Gomes, Jermann and Schmid (2014) examine a model with equilibrium default and nominally denominated console debt. They find that, so long as the expected maturity of debt exceeds one period, a temporary fall in inflation increases default rates and also generates a debt overhang channel affecting future investment and production decisions. Default in the Gomes, Jermann and Schmid economy triggers costly restructuring with no change in the number of firms. We instead associate default with exit. This aspect of our model combines with endogenous entry decisions to deliver a time-varying measure of producers. Furthermore, our model is also distinguished by persistent firm-level heterogeneity across periods. Because the distribution of firms enters into our aggregate state vector, there are persistent real effects from a purely financial shock despite the fact that debt is one-period lived.

Khan and Thomas (2013) study financial frictions in the form of collateralized borrowing. There, a credit shock is an unanticipated change in the fraction of their collateral firms can borrow against. When hit by such a shock, that model delivers a large, persistent recession with some similar features to the 2007 U.S. recession. Our current work departs from this study in three main ways. First, while capital serves as collateral, our borrowing terms depend on firm-level and aggregate state variables and arise from endogenous forward-looking lending schedules. Second, given the complexity associated with solving for equilibrium loan schedules, we omit the partial capital irreversibility considered there. Third, our model generates endogenous movements in entry and exit rates, and these series exhibit cyclical properties consistent with the U.S. data (see Campbell (1998)). We find this is important in generating more gradual recoveries in employment and investment following a shock affecting financial markets, and thus greater consistency with the post 2009Q2 U.S. experience.

Buera and Moll (2015) also study a heterogeneous agent model with borrowing subject to collateral constraints. In their setting, entrepreneurs have constant returns production, face i.i.d. productivity shocks and observe shock realizations a period in advance. As a result, the distribution in their model does not evolve gradually over time as in ours, and they find shocks to collateral constraints

are isomorphic to shocks to aggregate total factor productivity. Shourideh and Zetlin-Jones (2016) also study financial shocks in a model where heterogeneous firms face collateral constraints. Their model features two types of intermediate goods producers, publicly owned and privately owned. They argue that financial shocks are a promising source of aggregate fluctuations when there are strong linkages through intermediate goods trades across firms.

The financial frictions we study stem from non-contingent loans that introduce default into the model. This type of loan contract was first characterized by Eaton and Gersovitz's (1981) study of international lending. Aguiar and Gopinath (2006) and Arellano (2008) undertake quantitative analyses of sovereign debt. Chatterjee et al. (2008) study unsecured lending to households, and Nakajima and Rios-Rull (2005) extend the framework to include real aggregate shocks.

Our emphasis on productivity dispersion, non-contingent debt and equilibrium default is shared by Arellano, Bai and Kehoe (2012), who explore the extent to which aggregate fluctuations are explained by movements in the labor wedge driven by uncertainty shocks.⁴ Our study differs from theirs in that we explore aggregate responses to financial shocks, our employment levels are not predetermined, and we include capital investment. We share in common the simplifying assumption that defaulting firms must exit the economy.

The model we develop is consistent with the observation that the aggregate U.S. nonfinancial business sector can fully finance investment from cash flows. At the same time, changes in financial conditions have aggregate implications, given firms' differing reliance on external finance. On average, new firms begin with relatively small capital stocks. They grow gradually, maintaining their leverage in a narrow range. Conditional on survival, they ultimately achieve a capital level consistent with their expected productivity and begin reducing their debt, eventually changing its sign to build financial savings. In short, our firms have a natural maturing phase and tend to eventually outgrow default risk. Thus, the incidence of a credit shock differs, and we can explore the extent to which small firms are disproportionately affected. Following such shocks, shifts in the distribution of capital drive movements in aggregate total factor productivity through misallocation. In this respect, our study is also related to Buera and Shin (2013), who show that collateral constraints can protract the transition path to economic development if capital is initially misallocated.

⁴Gomes and Schmid (2014) develop a model with endogenous default where firms vary with respect to their leverage and study the implication for credit spreads. Credit spreads are also a focus of Gertler and Kiyotaki (2010), who study a model where such spreads are driven by agency problems arising with financial intermediaries.

3 Model

Our model has three types of economic agents: households, firms, and a perfectly competitive representative financial intermediary. Firms face persistent differences in their individual total factor productivities and their ability to borrow. Furthermore, their only source of external finance is one-period non-contingent debt provided by the financial intermediary at loan rates determined by their individual characteristics. These aspects of the model combine to yield substantial heterogeneity in production.

We assume a large number of firms. Each starts any period identified by its predetermined stock of capital, $k \in \mathbf{K} \subset \mathbf{R}_+$, the debt it took on in the previous period, $b \in \mathbf{B} \subset \mathbf{R}$, and its current firm-specific exogenous state, $\varepsilon \in \mathbf{E} \subset \mathbf{R}_+$. We summarize the distribution of firms over (k, b, ε) using the probability measure μ defined on the Borel algebra generated by the open subsets of the product space, $\mathbf{K} \times \mathbf{B} \times \mathbf{E}$. Its changes over time are shaped by firms' responses to persistent movements in both aggregate total factor productivity, z , and the economy's financial (or *credit*) state, ζ , affecting their financial positions and their borrowing terms.

Aggregate TFP is a Markov chain, $z \in \mathbf{Z} \equiv \{z_1, \dots, z_{N_z}\}$, with $\Pr(z' = z_g \mid z = z_f) \equiv \pi_{fg}^z \geq 0$ and $\sum_{g=1}^{N_z} \pi_{fg}^z = 1$ for each $f = 1, \dots, N_z$. Likewise, $\zeta \in \{\zeta_1, \dots, \zeta_{N_\zeta}\}$, with $\Pr\{\zeta' = \zeta_k \mid \zeta = \zeta_h\} \equiv \pi_{hk}^\zeta \geq 0$ and $\sum_{k=1}^{N_\zeta} \pi_{hk}^\zeta = 1$ for each $h = 1, \dots, N_\zeta$. The joint stochastic process is summarized by $s = (z, \zeta)$, and the underlying Markov chains, $\{\pi_{fg}^z\}$ and $\{\pi_{hk}^\zeta\}$, combine to deliver its transition probabilities, $\pi_{lm}^s \equiv \Pr(s' = (z, \zeta)_m \mid s = (z, \zeta)_l)$. We denote the real and financial shocks in each realized state s_l as z_l and ζ_l , for each $l = 1, \dots, N_s$, where $N_s = N_z N_\zeta$.

The aggregate state of the economy is fully summarized by (s, μ) . The distribution of firms therein evolves over time according to a mapping, Γ , from the current aggregate state; $\mu' = \Gamma(s, \mu)$. While taken as given by all agents, this evolution is determined in part by the actions of continuing firms and in part by entry and exit, as will be clear below.

3.1 Production, exit and entry

Each incumbent firm and each potential entrant (described below) can produce a homogenous output using predetermined capital stock k and labor n , via an increasing and concave production function; $y = z\tilde{\varepsilon}sF(k, n)$, where $F(k, n) = k^\alpha n^\nu$, with $\alpha > 0$, $\nu > 0$ and $\alpha + \nu < 1$. As noted above, z is stochastic total factor productivity common across firms, while $\tilde{\varepsilon}s$ is a firm-specific counterpart.

The firm-specific exogenous state, $\varepsilon \in \mathbf{E}$, has 3 components, the first two of which determine its productivity while the third determines whether or not it faces an earnings based borrowing limit.

Alongside productivity, these borrowing limits evolve stochastically. The first component of ε is a discrete-valued idiosyncratic component, $\tilde{\varepsilon} \in \{\tilde{\varepsilon}_1, \dots, \tilde{\varepsilon}_{N_{\tilde{\varepsilon}}}\}$, where $\Pr(\tilde{\varepsilon}' = \tilde{\varepsilon}_b \mid \tilde{\varepsilon} = \tilde{\varepsilon}_a) \equiv \pi_{ab}^{\tilde{\varepsilon}} \geq 0$, and $\sum_{b=1}^{N_{\tilde{\varepsilon}}} \pi_{ab}^{\tilde{\varepsilon}} = 1$ for each $a = 1, \dots, N_{\tilde{\varepsilon}}$.

The second is a permanent component drawn individually by each firm at its birth, $\tilde{s}$, which takes on one of $N_{\tilde{s}}$ values. This piece can be used to drive skewness in the firm sales distribution on the order of that in the U.S. data. The other component $\tilde{m} \in \{0, 1\}$ reflects whether or not the firm faces an earnings based borrowing limit in the present period. Each firm begins as an entrant subject to borrowing limits, $\tilde{m} = 1$. Thereafter, between each period and the next, it has probability φ_m of having these borrowing limits removed, $\tilde{m} = 0$, an absorbing state. Thus, the first row of the $\tilde{m}$ transition matrix is $[\varphi_m \ 1 - \varphi_m]$, and the second is $[0 \ 1]$. Let $\varepsilon = (\tilde{\varepsilon}, \tilde{s}, \tilde{m})$ summarize the firm-level productivity and borrowing limit, with transition matrix $\pi^\varepsilon = \{\pi_{ij}^\varepsilon\}$ derived from the underlying transition probabilities of its components, and $N_\varepsilon = 2N_{\tilde{\varepsilon}}N_{\tilde{s}}$ possible realizations. We represent the firm-level exogenous state in a given realization, ε_i , by $(\tilde{\varepsilon}_i, \tilde{s}_i, \tilde{m}_i)$ for convenience sake.

No microeconomic frictions impede capital reallocation in our model, so any firm's individual state can be effectively summarized by (x, ε) , where x is its cash on hand. Given real wage $\omega(s, \mu)$ and capital depreciation rate δ , the cash on hand of any type (k, b, ε) firm that produces is:

$$x(k, b, \varepsilon; s, \mu) = y(k, \varepsilon; s, \mu) - \omega(s, \mu) n(k, \varepsilon; s, \mu) + (1 - \delta)k - b - [\xi_0(\varepsilon) + \chi(\zeta) \xi_1(\varepsilon)],$$

where y and n represent the firm's chosen output and employment.⁵ Returning to our expression for cash on hand above, the final bracketed term is a fixed operating cost the firm must pay to maintain a place in the economy. This cost has two components, a state-invariant real resource cost, $\xi_0(\varepsilon)$, and a purely financial cost parameterized by $\xi_1(\varepsilon)$ present only in times of an aggregate credit shock; $\chi(\zeta)$ is an indicator function taking the value 1 when $s = (z, \zeta)$ is associated with a credit shock, and 0 otherwise. Both the real and financial fixed costs vary with firm productivity, hence are functions of ε .

A firm can evade its operating cost only by exiting the economy prior to production. If it chooses to do this, it forfeits its capital stock immediately and achieves a value of 0. As such, it necessarily defaults on its debt. Conversely, a firm failing to repay its debt must forfeit its capital stock and exit immediately, achieving value 0. Thus, a defaulting firm pays no operating cost.

Given our emphasis on studying how credit market frictions shape real economic outcomes, we impose a non-negativity restriction on firms' dividends to prevent them using equity to circumvent

⁵In what follows, both y and n vary with $\tilde{\varepsilon}$ and $\tilde{s}$, the first two components of ε that determine idiosyncratic productivity, but we write them as function of the whole vector for convenience.

frictions in debt markets. For the same reason, we impose some state-independent exit to prevent all of the economy's production taking place among firms that have effectively outgrown default risk and so always invest efficiently at borrowing costs never exceeding the equilibrium risk-free interest rate. In particular, we assume each firm faces a fixed probability $\pi_d \in (0, 1)$ that it will be forced to exit the economy after production in any given period. To maintain the number of firms in our economy at 1 on average, we also have exogenous birth of potential entrants.

At the start of any period, $\bar{\mu}^0$ potential entrants arrive with a common capital stock k_0 , idiosyncratic productivities $\tilde{\varepsilon}_0$ drawn from the ergodic distribution of $\tilde{\varepsilon}$ and financial positions b_0 drawn from a uniform distribution.

Each one becomes a new firm and starts producing immediately if it pays its operating cost, $\xi_0(\varepsilon) + \chi(\zeta)\xi_1(\varepsilon)$, and repays any start-up loan it has, $b_0 > 0$. Otherwise, it returns its capital to households and immediately leaves the economy.

3.2 Operation, investment and debt

Each firm takes a series of actions to maximize the expected discounted value of its dividends, given the aggregate state (s, μ) and its individual state (k, b, ε) . First, it decides whether to operate. To do so, it commits to repay any existing debt $b > 0$ and its operating cost $\xi_0(\varepsilon) + \chi(\zeta)\xi_1(\varepsilon)$ by the close of the period. Otherwise, it immediately exits the economy, forfeiting all revenues and capital.

Conditional on operating, a firm (k, b, ε) selects its current employment to solve: $\pi(k, \varepsilon; s, \mu) = \max_n [z\tilde{\varepsilon}sk^\alpha n^\nu - \omega(s, \mu)n]$, where z is given by $s = (z, \zeta)$. The resulting labor and production levels, alongside current profits, π , are listed below. Notice these choices are independent of debt, b , and so are common to all firms in operation that share a common (k, ε) .

$$n(k, \varepsilon; s, \mu) = [\nu z\tilde{\varepsilon}sk^\alpha / \omega(s, \mu)]^{\frac{1}{1-\nu}} \quad (1)$$

$$y(k, \varepsilon; s, \mu) = (\tilde{\varepsilon}sz)^{\frac{1}{1-\nu}} [\nu / \omega(s, \mu)]^{\frac{\nu}{1-\nu}} k^{\frac{\alpha}{1-\nu}} \quad (2)$$

$$\pi(k, \varepsilon; s, \mu) = (1 - \nu)y(k, \varepsilon; s, \mu) \quad (3)$$

Immediately after production, operating firms learn whether they can continue beyond the current period. Any firm drawing an exit shock takes on zero debt, sells its remaining capital and returns to its shareholders the value $\max\{x, 0\}$ as it leaves the economy.⁶ A continuing firm, by contrast, chooses its investment, current dividends, and the debt with which it will enter the next period.

⁶Our timing prevents the exit shock directly implying default. However, because operation decisions take place before its realization, some firms counting on new loans to cover part of their existing debt or operating costs may be

Conditional on operating and escaping the exit shock, a continuing firm determines its future capital, k' , future debt, b' , and current dividends, D . Its capital stock for the start of next period is:

$$k' = (1 - \delta) k + i, \quad (4)$$

where i is its investment, $\delta \in (0, 1)$ and primes denote one-period-ahead values. The firm may take on debt $b' > 0$ to finance any shortfall between its chosen capital and its cash on hand, x , the sum of its current profits and nondepreciated capital net of operating costs and existing debt repayment:

$$x = \pi(k, \varepsilon; s, \mu) + (1 - \delta) k - b - [\xi_0(\varepsilon) + \chi(\zeta) \xi_1(\varepsilon)]. \quad (5)$$

Given the default risk in our setting, firms' loan interest rates depend on their chosen debt and capital and their current productivities, as these attributes affect the distribution of future earnings and thus the probability the competitive lender will be repaid. For now, let $q(\cdot)$ summarize the loan discount factor faced by a borrowing firm. For each unit of debt it incurs for the next period, the firm receives $q(\cdot)$ units of output to use for investment or dividends. Thus a loan of $q(\cdot)b'$ implies debt of b' to be repaid next period, and the continuing firm's current dividends are $D(\cdot) = x - k' + q(\cdot)b'$.

3.3 Firm values

Firms selecting a common k' and b' will not all have the same income next period, given the firm-specific component of total factor productivity, $\tilde{\varepsilon}'\tilde{s}$. Among firms with the same (k', b') , those realizing high productivities may repay their loans, while others may default. If a firm defaults, the financial intermediary recovers a fraction, $\lambda(\zeta)$, of the firm's nondepreciated capital, with this fraction depending on current financial conditions. We assume the remainder of a defaulting firm's capital is lump-sum rebated to households, so that default implies no direct loss of resources.

When a continuing firm with current idiosyncratic productivity ε takes on a debt b' , alongside future capital k' , that firm receives a loan of size $q(k', b', \varepsilon; s, \mu) b'$. As noted above, the loan discount factor, $q(k', b', \varepsilon; s, \mu)$, is determined as a function of the firm's repayment probability. Absent any other frictions, competitive lending equates the financial intermediary's expected return on each of its loans to the risk-free real interest rate, $\frac{1}{q_0(s_l, \mu)} - 1$, where:

$$q_0(s_l, \mu) = \sum_{m=1}^{N_s} \pi_{lm}^s d_m(s_l, \mu), \quad (6)$$

caught short on finding they cannot continue to the next period. This nature-induced default is reflected in the values shareholders receive from such firms, $\max\{x, 0\}$, and in the determination of loan schedules below in section 3.4.

and $\pi_{lm}^s d_m(s_l, \mu)$ is the price of an Arrow security for each state $s' = s_m$. We assume the intermediary suffers a proportional cost $\tau(\zeta) \geq 1$ to take on risky debt, which implies a markup on the expected return it requires for such loans. Note that this risk markup depends on the aggregate financial state.

Firms' loan discount factors are bounded above by the risk-free factor $q_0(s, \mu)$ and below by 0. Given $\mu' = \Gamma(s, \mu)$ and its chosen (k', b') , the firm is offered $q(k', b', \varepsilon; s, \mu) < q_0(s, \mu)$ so long as there is any possible realization of (ε', s') that will lead it to default on its b' next period. Among firms selecting a common (k', b') , those realizing higher ε' will be less likely to default, as will be clear below. Thus, given persistence in firm productivities, $q(k', b', \varepsilon; s, \mu)$ is weakly rising in ε . For the same reasons, the firm's q rises in k' and falls in b' .

Recall the definition of an operating firm's cash on hand, x , from (5) above. There, we observed that a firm's individual levels of k and b do not separately determine any of its choices beyond their effect in x . To see this, note that a firm's resource constraint (determined by the non-negativity constraint on dividends) may be written as simply: $x - k' + q(k', b', \varepsilon; s, \mu) b' \geq 0$ and its earnings based borrowing limit is $q(k', b', \varepsilon; s, \mu) b' \leq \varphi(\varepsilon) x$. The latter constrains the choice of debt to not exceed $\varphi(\varepsilon)$ of its existing cash-on-hand. Firms chose $\varepsilon = (\tilde{\varepsilon}, \tilde{s}, \tilde{m})$ with $\tilde{m} = 0$ do not face an earnings based borrowing limit. In this case we simply assume that $\varphi(\tilde{\varepsilon}, \tilde{s}, 0)$ is sufficiently large that the borrowing limit never binds. Given the non-negativity condition on dividends and the possible borrowing limit, the firm's feasible capital and debt combinations are given by the set $\Phi(x, \varepsilon; s, \mu)$, where:

$$\Phi(x, \varepsilon; s, \mu) = \{(k', b') \in \mathbf{K} \times \mathbf{B} \mid x - k' + q(k', b', \varepsilon; s, \mu) b' \geq 0 \text{ and } q(k', b', \varepsilon; s, \mu) b' \leq \varphi(\varepsilon) x\}. \quad (7)$$

Since x fully captures previous decisions influencing its current choice set, the firm's value is a function only of x and ε . This important result allows us to reduce the firm-level state vector, and thus the dimension of the value and default functions that characterize competitive equilibrium.

Let $V(x, \varepsilon_i; s_l, \mu)$ represent the beginning of period value of a firm just before its default decision, and let $V^1(x, \varepsilon_i; s_l, \mu)$ represent its value conditional on operating. If $\Phi(x, \varepsilon_i; s_l, \mu) \neq \{\emptyset\}$, the firm can cover its operating cost and repay its debt while paying a non-negative dividend. In that case, the firm operates in the current period, achieving a non-negative value $V^1(x, \varepsilon_i; s_l, \mu)$. Otherwise, it defaults on its debt and immediately exits the economy with zero value,

$$V(x, \varepsilon_i; s_l, \mu) = \max\{V^1(x, \varepsilon_i; s_l, \mu), 0\}. \quad (8)$$

Given the constraint set in (7), it is straightforward to show that a firm's value is increasing in its cash on hand, x , and in its productivity, ε .

The value of operating, V^1 , accounts for the possibility that the firm will receive the exit shock after current production and be unable to continue to the next period. Recall that, with probability π_d , the firm is forced to exit at the end of the period. In that case, it simply returns its cash to shareholders as it exits, with the lender absorbing the loss if $x < 0$. Otherwise, it achieves the continuation value V^2 determined below.

$$V^1(x, \varepsilon_i; s_l, \mu) = \pi_d \max\{x, 0\} + (1 - \pi_d) V^2(x, \varepsilon_i; s_l, \mu) \quad (9)$$

Given the current aggregate state and $\mu' = \Gamma(s_l, \mu)$, continuing firms solve:

$$V^2(x, \varepsilon_i; s_l, \mu) = \max_{k', b'} \left[x - k' + q(k', b', \varepsilon_i; s_l, \mu) b' + \sum_{m=1}^{N_s} \pi_{lm}^s d_m(s_l, \mu) \sum_{j=1}^{N_\varepsilon} \pi_{ij}^\varepsilon V(x'_{jm}, \varepsilon_j; s_m, \mu') \right], \quad (10)$$

subject to :

$$(k', b') \in \Phi(x, \varepsilon_i; s_l, \mu) \quad (11)$$

$$x'_{jm} = \pi(k', \varepsilon_j; s_m, \mu') + (1 - \delta)k' - b' - [\xi_0 + \chi(\zeta_m) \xi_1(\varepsilon_j)], \quad (12)$$

where $V(\cdot)$ is defined in (8), $\Phi(\cdot)$ is given by (7), and $\pi(\cdot)$ is from (3).

3.4 Loan rates

Turning to the determination of loan discount factors, $q(\cdot)$, consider a firm entering next period with realized cash x' and productivity ε' , given aggregate state (s', μ') . Let $\chi(x', \varepsilon'; s', \mu')$ be an indicator taking the value 1 if the firm repays its debt, and 0 otherwise. Threshold values of cash on hand, $x^d(\varepsilon'; s', \mu')$, solve $V^1(x^d, \varepsilon'; s', \mu') = 0$ and separate those firms of a given productivity ε' for which $\chi(\cdot) = 1$ (those with $x \geq x^d$) from those for which $\chi(\cdot) = 0$ (those that default).

We have assumed that the financial intermediary providing loans to firms is perfectly competitive, so the interest rate it charges on any loan is determined by a zero expected profit condition. However, because the lender faces a proportional cost $\tau(\zeta)$ for risky debt on its balance sheets, it charges a markup for such loans. This is reflected by $\tau(k', b', \varepsilon_i; s_l, \mu)$, which takes the value $\tau(\zeta_l)$ for any loan not satisfying $\chi(x'_{jm}, \varepsilon_j; s_m, \mu') = 1$ for all realizable (ε_j, s_m) combinations, and is 1 otherwise.

Recall the determination of x'_{jm} from (12), and recall the possibility of a nature-induced default if an otherwise solvent firm with $x'_{jm} < 0$ draws an exit shock. In that event, the lender will recover whatever the firm has left after its operating costs, $\max\{x'_{jb} + b', 0\}$. Taking into account the fact that the intermediary recovers no more than $\lambda(\zeta)$ fraction of a firm's remaining capital in the event

of ordinary default, we arrive at the following implicit solution for the loan discount factor.

$$q(k', b', \varepsilon_i; s_l, \mu) b' \cdot \tau(k', b', \varepsilon_i; s_l, \mu) = \sum_{m=1}^{N_s} \pi_{lm}^s d_m(s_l, \mu) \sum_{j=1}^{N_\varepsilon} \pi_{ij}^\varepsilon \left[\chi(x'_{jm}, \varepsilon_j; s_m, \mu') \left[(1 - \pi_d) b' \right. \right. \\ \left. \left. + \pi_d \min\{b', \max\{x'_{jm} + b', 0\}\} \right] + \left[1 - \chi(x'_{jm}, \varepsilon_j; s_m, \mu') \right] \min\{b', \lambda(\zeta)(1 - \delta) k'\} \right] \quad (13)$$

Notice the loan price solving (13) gives the lender the same net expected return as that associated with risk-free assets, $q_0(s_l, \mu)^{-1}$. If a loan has no probability of default, then $\chi(x'_{jm}, \varepsilon_j; s_m, \mu') = 1$ and $x'_{jm} \geq 0$ for all (ε_j, s_m) with $\pi_{ij}^\varepsilon > 0$ and $\pi_{lm}^s(s_l) > 0$. In that case, $\tau(k', b', \varepsilon_i; \cdot) = 1$ and $q(k', b', \varepsilon_i; \cdot) b' = \sum_{m=1}^{N_s} \pi_{lm}^s d_m(s_l, \mu) b'$, so $\tau(k', b', \varepsilon_i; s_l, \mu) q(k', b', \varepsilon_i; s_l, \mu) = q_0(s_l, \mu)$.

3.5 Households

We close the model with a unit measure of identical households. Their wealth is held as shares in firms identified using the measure $\tilde{\lambda}$, and in one-period noncontingent bonds, ϕ .⁷ Given the cum-dividend prices of their current shares, $m_0(x, \varepsilon; s, \mu)$, the risk-free bond price $q_0(s, \mu)^{-1}$, and the real wage, $\omega(s, \mu)$, households select their current consumption, c , hours worked, n^h , new bond holdings ϕ' , and the numbers of new shares, $\tilde{\lambda}'(x', \varepsilon')$, to purchase at ex-dividend prices $m_1(x', \varepsilon'; s, \mu)$.

The lifetime expected utility maximization problem of the representative household is:

$$V^h(\tilde{\lambda}, \phi; s_l, \mu) = \max_{c, n^h, \phi', \tilde{\lambda}'} \left[U(c, 1 - n^h) + \beta \sum_{m=1}^{N_s} \pi_{lm}^s V^h(\tilde{\lambda}', \phi'; s_m, \mu') \right] \quad (14)$$

subject to

$$c + q_0(s, \mu) \phi' + \int m_1(x', \varepsilon'; s_l, \mu) \tilde{\lambda}'(d[x' \times \varepsilon']) \leq \left[\omega(s_l, \mu) n^h + \phi \right. \\ \left. + \int m_0(x, \varepsilon; s_l, \mu) \tilde{\lambda}(d[x \times \varepsilon]) \right]$$

and given $\mu' = \Gamma(s, \mu)$. Let $C^h(\tilde{\lambda}, \phi; s, \mu)$ and $N^h(\tilde{\lambda}, \phi; s, \mu)$ be the decision rules for consumption and hours worked. Let $\Phi^h(\tilde{\lambda}, \phi; s, \mu)$ be the decision rule for bonds, and let $\Lambda^h(x', \varepsilon', \tilde{\lambda}, \phi; s, \mu)$ be the shares purchased in firms that will realize next-period productivity ε' and cash x' .

4 Computing equilibrium

In *recursive competitive equilibrium*, each firm solves the problem described by (8) - (12), households solve the problem described in (14), loans are priced according to (13), the markets for labor,

⁷Households also have access to a complete set of state-contingent claims. However, absent household heterogeneity, these assets are in zero net supply in equilibrium, so we do not explicitly model them here.

output and firm shares clear, and the resulting individual decision rules for firms and households are consistent with the aggregate law of motion, Γ . Using $C(s, \mu)$ and $N(s, \mu)$ to describe the market-clearing values of household consumption and hours worked, it is straightforward to show that market-clearing requires that (a) the real wage equal the household marginal rate of substitution between leisure and consumption, (b) the risk-free bond price, q_0^{-1} , equal the expected gross real interest rate, and (c) firms' state-contingent discount factors be consistent with the household marginal rate of substitution between consumption across states.

$$\omega(s, \mu) = D_2 U\left(C(s, \mu), 1 - N(s, \mu)\right) / D_1 U\left(C(s, \mu), 1 - N(s, \mu)\right)$$

$$q_0(s, \mu) = \beta \sum_{m=1}^{N_s} \pi_{lm}^s D_1 U\left(C(s_m, \mu'), 1 - N(s_m, \mu')\right) / D_1 U\left(C(s, \mu), 1 - N(s, \mu)\right)$$

$$d_m(s, \mu) = \beta D_1 U\left(C(s_m, \mu'), 1 - N(s_m, \mu')\right) / D_1 U\left(C(s, \mu), 1 - N(s, \mu)\right).$$

We compute equilibrium in our economy by combining the firm-level optimization problem with the equilibrium implications of household utility maximization listed above, effectively subsuming households' decisions into the problems faced by firms. Without loss of generality, we assign $p(s, \mu)$ as an output price at which firms value current dividends and payments and correspondingly assume that firms discount their future values by the household subjective discount factor. Given this alternative means of expressing firms' discounting, the following three conditions ensure all markets clear in our economy.

$$p(s, \mu) = D_1 U\left(C(s, \mu), 1 - N(s, \mu)\right) \quad (15)$$

$$\omega(s, \mu) = D_2 U\left(C(s, \mu), 1 - N(s, \mu)\right) / p(s, \mu) \quad (16)$$

$$q_0(s, \mu) = \beta \sum_{m=1}^{N_s} \pi_{lm}^s p\left(s_m, \Gamma(s, \mu)\right) / p(s, \mu) \quad (17)$$

We reformulate (8) - (12) here to obtain an equivalent, more convenient, representation of the firm problem with each firm's value measured in units of marginal utility, rather than output.

$$v(x, \varepsilon_i; s_l, \mu) = \max\left\{v^1(x, \varepsilon_i; s_l, \mu), 0\right\}. \quad (18)$$

$$v^1(x, \varepsilon_i; s_l, \mu) = \pi_d p(s_l, \mu) \max\{x, 0\} + (1 - \pi_d) v^2(x, \varepsilon_i; s_l, \mu) \quad (19)$$

$$\begin{aligned}
v^2(x, \varepsilon_i; s_l, \mu) = & \max_{k', b'} \left(p(s_l, \mu) [x - k' + q(k', b', \varepsilon_i; s_l, \mu) b'] \right. \\
& \left. + \beta \sum_{m=1}^{N_s} \pi_{lm}^s \sum_{j=1}^{N_\varepsilon} \pi_{ij}^\varepsilon v(x'_{jm}, \varepsilon_j; s_m, \mu') \right), \text{ subject to (11) - (12).}
\end{aligned} \tag{20}$$

The problem listed in equations (18) - (20) forms the basis for solving equilibrium allocations in our economy, so long as the prices p, ω and q_0 taken as given by firms satisfy the restrictions in (15) - (17), and the loan price schedules offered satisfy (13).

As noted above, a firm of type (k, b, ε) hires labor and produces only if $\Phi(x, \varepsilon_i; s_l, \mu) \neq \{\emptyset\}$ and $v^1(x, \varepsilon; s, \mu) \geq 0$, with x defined in (5). In that case, its decision rules for labor and output are given by (1) - (2), and its flow profits are given by (3). The more challenging objects we must determine are D, k' and b' for firms continuing into the next period. These decisions are dynamic, inter-related functions of firm productivity, ε , and cash on hand, x .

To solve the problems of continuing firms, we use a partitioning analogous to that in Khan and Thomas (2013), here extended for the fact that we study noncontingent debt with default and exit. In particular, we assign firms across three distinct categories reflecting the extent to which their investment activities can be affected by financial frictions and identify their decision rules accordingly. Firms termed (financially) *unconstrained* are those that have permanently outgrown the implications of financial frictions.

Firms that are (financially) *constrained type 1* can undertake efficient investment in the current period while borrowing at the risk-free interest rate, but face the possibility of paying a risk premium in future. *constrained type 2* firms cannot finance efficient investment without immediate default risk and thus a risk premium.

4.1 Decisions among unconstrained firms

By definition, an unconstrained firm has accumulated sufficient cash on hand that it can finance its efficient investments with no risk of default in every possible future state over the remainder of its lifetime. Any such firm has effectively outgrown financial frictions, so its marginal valuation on retained earnings equals the household marginal valuation of consumption, p , and it is indifferent between dividends and financial savings, $b' < 0$. Viewed another way, the firm's value function is linear in its debt or financial savings, so its k' decision is unaffected by its b' . Unless an exit shock prevents its continuation, each such firm adopts its efficient capital stock $k^*(\varepsilon_i; s_l, \mu)$ solving (21), achieving value $w^2(\cdot)$, and, given its financial indifference, it is content to adopt the debt policy

$B^w(\cdot)$ we isolate below to maintain that indifference permanently.

$$w^2(x, \varepsilon_i; s_l, \mu) = \max_{k'} \left[p(s_l, \mu) [x - k' + q_0(s_l, \mu) B^w(\varepsilon_i; s_l, \mu)] \right. \\ \left. + \beta \sum_{m=1}^{N_s} \pi_{lm}^s \sum_{j=1}^{N_\varepsilon} \pi_{ij}^\varepsilon w^0(x'_{jm}, \varepsilon_j; s_m, \mu') \right], \quad (21)$$

where x'_{jm} is given by (12) and $w^0(x, \varepsilon; s, \mu) = p(s, \mu) \pi_d \max\{x, 0\} + (1 - \pi_d) w^2(x, \varepsilon; s, \mu)$.

We assign an unconstrained firm a *minimum savings* debt policy solving (22) - (24) to ensure that it always maintains sufficient wealth to implement its optimal investments with no default risk under all possible future paths of (ε, s) .⁸ Let $\tilde{B}(k', \varepsilon_j; s_m, \Gamma(s, \mu))$ define the maximum debt level at which a firm entering next period with k' and realizing (ε_j, s_m) will be unconstrained. This requires that the firm can adopt $k^*(\varepsilon_j; s_m, \mu')$ and $B^w(\varepsilon_j; s_m, \mu')$ next period while maintaining $D \geq 0$, and that $x'_{jm} \geq 0$ so an exit shock cannot lead it to default.⁹ The unconstrained firm debt policy, $B^w(\varepsilon_i; s_l, \mu)$, is the minimum $\tilde{B}_{jm}$; i.e., it is the maximum debt with which the firm can exit this period and be certain to be unconstrained next period, given that it adopts $k^*(\varepsilon_i; s_l, \mu)$.

$$B^w(\varepsilon_i; s_l, \mu) = \min_{\{\varepsilon_j | \pi_{ij}^\varepsilon > 0 \text{ and } s_m | \pi_{lm}^s > 0\}} \tilde{B}(k^*(\varepsilon_i; s_l, \mu), \varepsilon_j; s_m, \Gamma(s_l, \mu)) \text{ where} \quad (22)$$

$$\tilde{B}_1(k, \varepsilon_i; s_l, \mu) = e(k, \varepsilon_i; s_l, \mu) + \min \left\{ \left[q_0(s_l, \mu) B^w(\varepsilon_i; s_l, \mu) - k^*(\varepsilon_i; s_l, \mu) \right], 0 \right\}, \quad (23)$$

$$\tilde{B}(k, \varepsilon_i; s_l, \mu) = \min \left\{ \tilde{B}_1(k, \varepsilon_i; s_l, \mu), e(k, \varepsilon_i; s_l, \mu) - \frac{q_0(s_l, \mu) B^w(\varepsilon_i; s_l, \mu)}{\varphi(\varepsilon_i)} \right\}, \quad (24)$$

$$\text{and } e(k, \varepsilon_i; s_l, \mu) = \pi(k, \varepsilon_i; s_l, \mu) + (1 - \delta)k - [\xi_0 + \chi(\zeta_l) \xi_1(\varepsilon_i)]. \quad (25)$$

In (22) we ensure that $B^w(\varepsilon_i; s_l, \mu)$ is small enough that the firm can afford $k^*(\varepsilon_i; s_l, \mu)$ this period for any outcome $\varepsilon_i; s_l$ and that it satisfies any borrowing limit the firm might face. Given their capital and debt decision rules, the dividends paid by continuing unconstrained firms are:

$$D^w(x, \varepsilon_i; s_l, \mu) = x - k^*(\varepsilon_i; s_l, \mu) + q_0(s_l, \mu) B^w(\varepsilon_i; s_l, \mu). \quad (26)$$

4.2 Decisions among constrained firms

For any continuing firm (x, ε) not yet classified as unconstrained, we first consider whether it has crossed the relevant threshold to have become so. This may be confirmed using equation 26 with the unconstrained firm decision rules (21) - (24). If $D^w(x, \varepsilon; s, \mu) \geq 0$, the firm (x, ε) is unconstrained

⁸We adopt this policy rather than a zero-dividend policy so as to bound the financial savings of long-lived firms.

⁹The second condition stops the firm defaulting by choice, since $v^1(x^d, \varepsilon_i; s_l, \mu) = 0$ is only satisfied by $x^d \leq 0$. Thus, $\tilde{B}_{il}$ must satisfy simply $\Phi_{il} - \tilde{B}_{il} \geq 0$ and $\Phi_{il} - \tilde{B}_{il} + q_0 B_{il}^w - k_{il}^* \geq 0$, where $\Phi_{il} = \pi_{il} + (1 - \delta)k - [\xi_0 + \chi_l \xi_{1,i}]$.

and those decision rules apply. Thus, the firm is unconstrained if its cash satisfies $x \geq x^{**}(\varepsilon; s, \mu)$, where:

$$x^{**}(\varepsilon; s, \mu) \equiv k^*(\varepsilon; s, \mu) - q_0(s, \mu)B^w(\varepsilon; s, \mu).$$

Otherwise, the firm is still financially constrained in that its available cash may affect its investment decisions now or in future, so its choices of capital and debt remain intertwined. Constrained firms of type 1 can invest to the efficient capital $k^*(\varepsilon; s, \mu)$ from (21) this period while paying no risk premium, as their next-period cash distribution implies zero probability of endogenous or nature-induced default; $\chi(x'_{jm}, \varepsilon_j; s_m, \mu') = 1$ and $x'_{jm} \geq 0$ for all j, m such that $\pi_{ij}^\varepsilon > 0$ and $\pi_{lm}^s > 0$, where x'_{jm} is from (12). In addition the level of borrowing this involves, $q_0(s, \mu)b' \leq \varphi(\varepsilon)x$. Such firms adopt $k^*(\varepsilon; s, \mu)$ and borrow at the risk-free rate. Because they are sure to remain in the economy through the next period, but they have not permanently outgrown financial frictions, their shadow value of internal funds exceeds the valuation on dividends. Thus, they set $D = 0$ and $q_0(s, \mu)b' = \min\{[k^*(\varepsilon; s, \mu) - x], \varphi(\varepsilon)x\}$.

Recall from section 3.4 that the default thresholds $x^d(\varepsilon)$ below which $\chi(x, \varepsilon; s, \mu) = 0$ are those where there is zero value of operating:

$$\pi_d \max\{x^d, 0\} + (1 - \pi_d) \max_{k', b'} \left[p(\cdot) [x^d - k' + q(k', b', \varepsilon; \cdot) b'] + \beta \sum_{m=1}^{N_s} \sum_{j=1}^{N_\varepsilon} \pi_{lm}^s \pi_{ij}^\varepsilon v(x'_{jm}, \varepsilon_j; \cdot) \right] = 0,$$

so $x^d(\varepsilon) \leq 0$ always. Hence, if a firm has zero probability of realizing $x'_{jm} < 0$ next period, it also has zero probability of a start-of-period default. This means that an constrained firm (x, ε) is type 1 if the following inequality is satisfied for all possible j, m outcomes.

$$\pi(k^*(\varepsilon; s, \mu), \varepsilon_j; s_m, \Gamma(s, \mu)) + (1 - \delta)k^*(\varepsilon; s, \mu) - [\xi_0 + \chi(\zeta_m)\xi_1] - q_0(s, \mu)^{-1}[k^*(\varepsilon; s, \mu) - x] \geq 0$$

For constrained firms that do not satisfy the type 1 check above, we know of no convenient way to separate the loan implied by a given k' choice to distinctly identify the corresponding debt level, b' . Unlike type 1 firms, a type 2 constrained firm may find $D = 0$ suboptimal; it may in fact achieve higher expected discounted value by paying dividends in the current period if it faces sufficiently high probability that it will be forced to default and exit the economy at the start of the next period. Thus, we must isolate the k' , b' and D choices of any such firm by directly solving the problem listed in (18) - (20), which we do using a two dimensional grid on k' and b' .

The financial frictions in our economy yield two types of misallocation reducing aggregate TFP, both transmitted through the decisions of type 2 constrained firms. First, these firms adopt inefficiently small capital stocks because they either cannot borrow to k^* or are unwilling to suffer the

required risk premium to do so. Second, their low cash on hand and poor financing terms may lead them to default on their loans and exit the economy. Given decreasing returns to scale, the resulting loss of a production unit further distorts the allocation of aggregate capital away from the efficient one.

5 Calibration

We explore the firm-level and aggregate implications of imperfect credit markets across a series of numerical exercises below. We assume that the representative household's period utility is the result of indivisible labor (Rogerson (1988)): $u(c, L) = \log c + \varphi L$. The firm-level production function is Cobb-Douglas: $z\varepsilon F(k, n) = z\varepsilon k^\alpha n^\nu$.

We set the length of a period to be one year and select parameters governing preferences and aspects of technology common across firms as follows.¹⁰ First, we set the household discount factor, β , to imply an average real interest rate of 4 percent, consistent with findings by Gomme, Ravikumar and Rupert (2008). Next, the production parameter ν is set to yield an average labor share of income at 0.60 (Cooley and Prescott (1995)). The depreciation rate, δ , is taken to imply an average investment-to-capital ratio of roughly 0.069, matching the average value for the private capital stock between 1954 and 2002 in the U.S. Fixed Asset Tables, controlling for growth. Given this value, we determine capital's share, α , so that our model matches the average private capital-to-output ratio over the same period, at 2.3, and we set the parameter governing the preference for leisure, φ , to imply an average of one-third of available time is spent in market work.

Exact aggregation obtains in an otherwise identical reference model without financial frictions. We use that model to estimate an exogenous stochastic process for aggregate productivity. We begin by assuming the shock follows a mean zero AR(1) process in logs: $\log z' = \rho_z \log z + \eta'_z$ with $\eta'_z \sim N(0, \sigma_{\eta_z}^2)$. Next, we estimate the values of ρ_z and σ_{η_z} from Solow residuals measured using data on real U.S. GDP and private capital, together with the total employment hours series constructed by Cociuba, Prescott and Ueberfeldt (2012) from CPS household survey data, over the years 1959-2012. Then, we discretize this process as a 3-state Markov Chain; $N_z = 3$.

As noted above, our exogenous aggregate state also includes credit shocks. These shocks, ζ , follow a 2-state Markov chain with realizations $\{\zeta_o, \zeta_l\}$ and the transition matrix below. We associate ζ_o with ordinary credit conditions and ζ_l with a credit shock (low credit conditions). We assume $\zeta = \zeta_o$ in calibrating our steady state and assume no financial component of operating costs there,

¹⁰Our annual calibration allows us to be consistent with establishment-level firm size data.

$\xi_1(\varepsilon_i) = 0$. In times of ordinary credit, the capital recovery rate λ_o is selected to imply a 43.4 percent debt recovery rate consistent with Moody's debt recovery rates over 1982-2010, stripping away years associated with the S&L crisis and the recent financial crisis.

$$\Pi^\zeta = \begin{bmatrix} p_o & 1 - p_o \\ 1 - p_l & p_l \end{bmatrix}$$

The probability of continuing in ordinary borrowing conditions is p_o , $\Pr\{\zeta' = \zeta_o | \zeta = \zeta_o\}$, while $1 - p_l$ is the probability of escape from crisis conditions, $\Pr\{\zeta' = \zeta_o | \zeta = \zeta_l\}$. We select the parameters of the Π^ζ matrix using evidence on banking crises from Reinhart and Rogoff (2009). Their definition of a banking crisis includes episodes where bank runs lead to the closure or public takeover of financial institutions as well as those without bank runs where the closure, merging, takeover or government bailout of one important financial institution is followed by similar outcomes for others. They document 13 crises in the U.S. since 1800 and the share of years spent in crises at 13 percent, which together imply an average crisis duration of 2.09 years. Given our use of postwar targets to calibrate the remaining parameters of our model, the more appropriate statistics for our purposes are those from the period 1945-2008, wherein the U.S. has had two banking crises (the 1989 savings and loan crisis and the 2007 subprime lending crisis).¹¹ Unfortunately, it is not possible to determine the average length of a U.S. crisis from this sample period, without knowing the ending date of the most recent crisis. Given this difficulty, alongside Reinhart and Rogoff's argument that the incidence and number of crises is similar across the extensive set of countries they consider, we focus instead on their data for advanced economies. The average number of banking crises across advanced economies over 1945 - 2008 was 1.4, while the share of years spent in crisis was 7 percent. Combining these observations, we set $p_o = 0.9765$ and $1 - p_l = 0.3125$ so that the average duration of a credit crisis in our model is 3.2 years, and the economy spends 7 percent of time in the crisis state.

The growth of young firms is closely tied to the costs of loans as well as their earnings based borrowing constraints. Furthermore, we emphasise the size-over-age distribution of firms as this governs how quickly firms grow which, in turn is related to their dependence of loans and the maturing phase evident in the firm age-size distribution versus the skewness of the unconditional size distribution.

We calibrate parameters governing the distribution of firms over productivity, capital and financial assets to several aggregate financial moments including nonfinancial leverage and default rates as well

¹¹These observations are consistent with findings by Bianchi and Mendoza (2012); they document a frequency of financial crisis at 3 percent, consistent with three financial crisis in the U.S. over the past hundred years. Mendoza (2010) estimates a crisis frequency of 3.6 percent across emerging economies since 1980.

as the indebtedness of entrants. In the absence of comprehensive financial data on firms' balance sheets, unavailable for the majority of firms that are privately held, we target the size-over-age distribution. As our model is consistent with the relative size of entrants, the subsequent growth of young firms is closely related to their ability to finance investment. As young firms have low levels of retained earnings, their investment rates are sensitive to the cost of borrowing. Thus, the size-over-age distribution is highly correlated with the distribution of firms over debt, and lack of direct evidence of the latter may be partly overcome by emphasising our model's consistency with the former.

We will use a financial shock sufficient to drive a fall in measured TFP that is comparable to that observed over the 2007-9 U.S. recession. The steepness of the downturn and gradualism of recovery are critically affected by which misallocation channel (intensive- or extensive-margin) is dominant in hastening recession, and that this in turn depends on the shape of the average firm age-size distribution. This average distribution in fact determines the aggregate elasticity of response to credit conditions and the size of a financial shock required to reproduce the observed fall in real variables would differ across different models varying in their firm size distributions.

A credit shock ($\zeta = \zeta_l$) in our baseline model halves the default recovery rate, from $\lambda_l = 0.42$ to 0.21 and triggers financial operating costs, $\xi_1(\varepsilon_i) = 0.01\pi^*(k^*(\varepsilon_i), \varepsilon_i)$, where $\pi^*(k^*(\varepsilon_i), \varepsilon_i)$ is the steady state profit of a firm operating with the efficient level of capital for ε_i . Most importantly, it introduces a loan rate wedge $\tau(k', b', \varepsilon_i; \cdot) = (1.02)^{-1}$ raising the cost of any loan, $b > 0$, by 2 percent. These increases in the cost of borrowing and reductions in firms' cash-on-hand generates an endogenous decline in measured TFP in our model that is approximately 80 percent of the exogenous TFP shock we separately consider below toward comparing quantity responses to real versus financial shocks.

We allow the cash shock, ξ_1 , to vary in ε_i in an effort to even the incidence of the balance sheet shock across firms of differing sizes. Note that $\xi_1(\varepsilon_i)$ is termed a balance sheet shock because it applies only when $\zeta = \zeta_l$, and it is a purely financial shock redistributing cash from firms to households. To be clear, in contrast to the real operating cost (ξ_0) paid by firms in every period, $\xi_1(\varepsilon_i)$ does not enter the aggregate resource constraint. Thus, recalling that defaulting firms' capital not recovered by lenders is always lump-sum rebated to households, a credit shock has no direct consequence for real resources. The fall in recovery rates has minimal effect in our model, while the balance sheet shock increases default thereby reducing the number of operating firms. In contrast to this extensive margin effect on the distribution of firms, the loan rate shock is an intensive

margin effect that raises the costs of borrowing for all firms.

The distribution of idiosyncratic productivity for potential new firms is consistent with the invariant distribution of ε . These firms have common initial level of capital, k_0 , but draw an initial level of debt from a Pareto distribution with bounds $(\underline{b}_0, \bar{b}_0)$. We adopt the common k_0 choice to reduce the dimension of the distribution of entrants and have it vary over ε and b . The level of debt is set to reproduce the average level of indebtedness of entrants reported in the data discussed below. When initial debt does not lead to default, productivity and initial capital determine the relative size of young firms. Subsequently, debt, given productivity and initial capital, determines cash-on-hand and thus a firm's ability to finance investment. The evolving distribution of capital and productivity determines relative employment shares among young firms.

In the following results, focusing on the size-over-age distribution and not the unconditional distribution of firms over employment, we abstract from permanent differences in productivity and set $N_{\tilde{s}} = 1$. All firms draw productivity from a log-normal distribution with persistence ρ_ε and standard deviation σ_ε . We discretize $\tilde{\varepsilon}$ using 7 values, firms also vary in $\tilde{m}$ which is 0 for those borrowing limited by their earnings. Thus there are 14 values to $\varepsilon = \{\tilde{\varepsilon}, \tilde{m}\}$. Table 1 lists our aggregate parameters and several parameters that affect the distribution of firms over productivity, capital and debt.

TABLE 1. Parameter Values

(a) aggregate parameters								
β	φ	α	ν	δ	ρ_z	σ_{η_z}	p_o	p_l
0.96	2.5	0.26	0.60	0.0695	0.909	0.015	0.977	0.687
(b) firm distribution parameters								
π_d	$\bar{\mu}_0$	k_0	$\underline{b}_0$	$\bar{b}_0$	λ	ρ_ε	σ_ε	
0.035	0.19	0.1	-0.049	0.351	0.42	0.65	0.035	

Having chosen the first 5 parameters in Table 1 to match long-run aggregate moments, we adopt Syverson's estimated persistence for firms' productivities. Given other parameter values, we set the measure of potential new firms each period, $\bar{\mu}_0 = 0.1898$, so that our long-run average number of firms in production is 1. Our remaining parameters are chosen jointly to reproduce the average debt recovery rate, the aggregate debt-to-asset ratio and a series of moments from the Business Dynamics Statistics database (BDS) reflecting the average U.S. firm age-size distribution that are listed in Tables 3 and 4.

We select k_0 (lower bound for capital in new firms), $\underline{b}_0$ (curvature parameter for new firm capital distribution), $\bar{b}_0$ (a common level of debt for potential entrants), π_d (exogenous departure rate among producing firms), λ (fraction of capital stock recovered under default in times of ordinary credit) and σ_ε^2 (normal firms' idiosyncratic productivity variance) to reproduce the following empirical targets drawn from the BDS except where otherwise noted. Our targets are: (1) the average debt-to-asset ratio of nonfarm nonfinancial businesses (0.372; Flow of Funds 1954-2006), (2) average debt recovery rates under default in non-crisis years (0.434; Moodys Analytics), (3) unconditional average firm exit rate (8.65 percent), (4) average debt-to-asset ratio among entrants (40.0 percent; Kauffman Firm Survey), (5) average employment (size) among entrants, relative to incumbent firms (28.5 percent), (6) employment share of firms aged 0 - 5 years (17.0 percent), (7) survival rates through age 5 (45.4 percent), and (8-10) firm exit rates in years 1, 2 and 3, respectively (21.3, 15.9 and 13.4 percent).

Additional parameters are chosen to reproduce the age over size distribution and survival rates shown in Tables 3 and 4 below. Starting with our earnings based borrowing limits, we assume that all entrants start with $\tilde{m} = 1$, and have a constant probability of it disappearing, that it permanently moving to a state where they never again face borrowing limits. We find that we need this probability to be 0.025 to reproduce the observed growth rate of firms in the data. Borrowing limits themselves vary with ε , specifically a firms' idiosyncratic total factor productivity. Over the 7 level of productivity, $\varphi \in \{0.75, 0.75, 0.38, 0.38, 0.38, 0.15, 0.15\}$ where the first term represents the borrowing limits as a fraction of cash-on-hand, for firms with the lowest level of productivity, $\tilde{\varepsilon}_1$, and the last element is the corresponding value for firms with $\tilde{\varepsilon}_7$. We see that more productive firms, with higher target levels of capital for next period, experience more severe borrowing limits as a fraction of their cash-on-hand.

As discussed above, our borrowing limits are essential in reproducing the size over age distribution without assuming high levels of default risk to drive high loan rates. The risk of default, in our model, comes from the fixed costs of operating. We assume that $\xi_0(\varepsilon) = \xi_0^1 + \xi_0^2 \pi^*(k^*(\varepsilon_i), \varepsilon_i) + \xi_0^3 (\pi^*(k^*(\varepsilon_i), \varepsilon_i))^2$ where, as above, $\pi^*(k^*(\varepsilon_i), \varepsilon_i)$ describes steady state profits for a firm with $\tilde{\varepsilon}_i$ as the first component of ε_i . We set $\{\xi_0, \xi_1, \xi_2\} = \{0.01, 0.005, 0.015\}$.

6 Results

We begin this section with a summary of our model's steady state. Next, we present moments drawn from a long simulation with real and financial shocks. Thereafter, we compare our results to the Great Recession.

Starting with the long run, hindrances to borrowing arise endogenously in our model and are backward- *and forward*-looking. Among any group of firms with the same cash, irrespective of their current capital, a firm with higher current productivity obtain better loan rates, for any level of leverage, than another with lower productivity. Compared to settings with exogenous borrowing limits or collateral constraints, allowing productivity to matter in loan arrangements tends to reduce capital misallocation across firms. This creates a tension in efforts to reproduce the slow maturing phase evident in the U.S. firm age-size distribution (see Table 2). In the table, all results are reported as shares of employment of average incumbent firms. The top row reports the size of entrants and firms aged 1 to 5 years. Even 5 years after their entry, the typical young firm is less than half the size of the typical firm.

TABLE 2. Employment and Population Shares by Age

age	0	1	2	3	4	5
U.S. Data 1988-2006	28.46	37.07	40.39	43.55	46.63	49.67
Model	27.24	27.16	28.33	47.15	56.68	66.13

Models where young, small firms borrow subject to loan rates that are risk-based, and face no other barrier to growth, have difficulty, in reproducing the slow growth seen below in Table. If the costs of borrowing are determined by default risk, then failure rates must be high to imply loan rates large enough to slow growth to the rates seen in the data. However in models such as ours, high failure rates require large fixed costs of operation to reduce the level of earnings relative to any sized loan. The implication of such large costs is high failure. But the required high rates of exit would be inconsistent with the data on survival rates for a cohort. In Table 3 we see that almost half of a cohort of young firms survive to age 5, in the data.

TABLE 3. Cohort Survival Rates by Age

age	0	1	2	3	4	5
U.S. Data 1988-2006	100.0	78.76	66.29	57.42	50.73	45.61
Model	100.0	76.45	61.45	53.82	48.69	44.68

We resolve this tension between slow growth and moderate failure rates by increasing the extent to which a firms' ability to borrow is backward-looking. Assuming that all firms enter the economy subject to earnings based borrowing limits, and that these are very persistent, firms are unable to finance the levels of investment they would given loan rates. Further, we assume the fraction of earnings that a firm may borrow varies with its productivity. More productivity firms, the very

firms that would borrow at the lowest rates given the persistence of productivity, face the harshest borrowing limits in our model. While firms outgrow borrowing limits at the same rate, regardless of their size or productivity, the probability is small, 2.5 percent, and the expected duration of earnings based loan limits is long.

The importance of earnings based borrowing limits, in our quantitative model, implies that slow growth of young firms comes very naturally in a settings with exogenous collateral constraints. While in recent years have seen an emphasis on the difference between these two types of borrowing limits, relative to loan rates that are functions of the risk of default, the two types of borrowing limits are both largely backward looking. However to the extent earnings move with productivity, and productivity is persistent, earnings based borrowing limits do facilitate borrowing for more productive firms in the same way loan rates are lower for such firms. Nonetheless, in the absence of borrowing limits, our model with forward-looking loan rate determination, plus selection effects in firms' exit decisions and absent real frictions, would find it challenging to reproduce the size over age distribution of firms in the data. Nonetheless, given any long-run distribution of firms, default risk can amplify business cycles when compared to models with exogenous borrowing limits. We will show below that shocks to financial conditions cause large reductions in real economic activity in our model. This is in part because, at least in our model, default leads to procyclical entry and countercyclical exit, which themselves amplify the effects of shocks. All said and done, the more closely the model fits the age distribution, the greater the average misallocation in our model, and thus the rise in incumbent misallocation following a credit shock.

All this aside, the aggregate implications of financial frictions in our model are considerable even in steady state. In Table 4 we see the division of firms over financial frictions. Unconstrained firms, recall, face no difficulty in financing efficient levels of investment now or in the future. Constrained type 1 firms emulate the former set today, but have a positive probability of being unable to borrow at a risk free rate to adopt the efficient level of capital, in some future state. Constrained type 2 firms finance loans with risk premia in the current period, and thus underinvest relative to their productivity in the sense that firms with high cash on hand would choose a larger capital stock for next period. Constrained type

Figure 1: Steady state production time distribution over ϵ

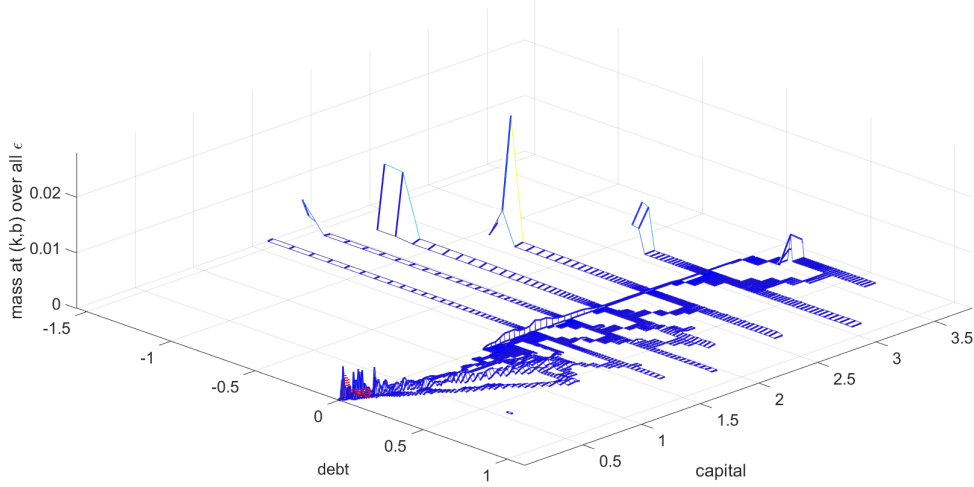


TABLE 4. Percent Population Shares by Type

<i>firm type</i>	<i>decisions</i>	<i>baseline case</i>
<i>constrained 2 (default)</i>	immediate exit	13.41
<i>constrained 2 (continue)</i>	(k', b') affected by x	47.75
<i>constrained 1</i>	$k^*(\varepsilon)$ and $b' = \frac{k^*(\varepsilon) - x}{q_0}$	5.00
<i>unconstrained</i>	$k^*(\varepsilon)$ and $b' = B^w(\varepsilon)$	33.5

Table 4 summarizes the distribution of cash-on-hand for firms at the start of the period, as many of these firms do not go on to produce, the table does not correspond to a distribution of firm observed in production. The top two rows are cash-poor, constrained type-2 firms that have not yet reached their efficient capital stocks. These firms' investments depend on their cash, and they have an immediate risk of default. The group in row 1 that would have cash on hand below 0 if they operated are type 2 firms that default and immediately exit.¹² Any firms in the next row are type 2 firms that produce and enter the coming period with capital levels influenced by their cash.

When we look at the distribution of types in production, the steady state has type-0, type-1 and type-2 firms that are 39.09, 5.77 and 55.14 percent of the unit measure of firms in production. The

¹²Threshold cash levels required to prevent default are negative at higher productivity realizations.

heterogeneity in firm types in the steady state lead to an average coefficient of variation for capital, across productivity types, of 1.0. Relative to an otherwise identical economy with perfect credit markets, the allocation of capital is distorted by the fact that firms' choices of future capital depend upon their cash on hand. Absent financial frictions, each firm with current productivity ε_i would simply select k' to equate the unit purchase price of investment to its expected discounted marginal return to capital, $q_0 \sum \pi_{ij}^\varepsilon [\pi_k(k', \varepsilon_j) + 1 - \delta]$. Thus, k' would depend only upon ε_i , and the coefficient of variation of firms' capital choices, conditional on ε , would be 0. Alongside this intensive margin misallocation among continuing firms, realized default generates extensive margin misallocation in the form of lost firms and their capital sitting idle while that exit takes place.

Examining firms across different levels of productivity, all facing earnings based borrowing limits, the coefficient of variation ranges from 0.89 to 1.40, increasing in productivity as we assume more severe borrowing limits for more productive firms. When examining the same range over productivity, for the group of firms that has outgrown borrowing limits, the corresponding values range from 0.23 to 0.25 with no systematic pattern over idiosyncratic productivity. We see that earnings based borrowing limits substantially increase the differences in capital held by firms with the same productivity, and thus the misallocation of capital in the steady state. This leads to an aggregate total factor productivity of 0.845. Aggregate capital before production is 1.45, as some of this is held by firms that default at the start of the period, that in production is 1.42. The default rate is 5.27 which, with exogenous exit shocks of 0.035 implies an overall exit rate of 7.9, close to the 8.65 seen in the BDS. While our default rate is much higher than that associated with corporate bonds, our distribution of firms represents that of the economy and not merely public firms with relatively low costs of borrowing.

6.1 Business cycles

We next consider business cycles in our model. We solve for stochastic equilibrium using the method of Krusell and Smith (1998), approximating the endogenous component of the aggregate state using the first moment of the distribution of capital. Table 5 reports the results of a 10,000 period simulation of our full baseline model, where business cycles are driven by shocks to total factor productivity and by financial shocks.

TABLE 5. Baseline Business Cycles with TFP Shocks and Credit Shocks

$x =$	GDP	C	I	N	K	r
$mean(x)$	0.501	0.390	0.112	0.338	1.244	0.042
σ_x/σ_{GDP}	(2.402)	0.442	3.802	0.746	0.574	0.351
$corr(x, GDP)$	1.000	0.721	0.915	0.913	0.135	0.729

Results of 10,000 period simulation. Columns: GDP, consumption, investment, hours worked, capital and risk-free real interest rate. Rows 2 and 3 report second moments for HP-filtered series using weight 100; $corr(exit, GDP) = -0.159$; $corr(entry, GDP) = 0.135$.

Given our section 5 calibration, credit shocks are rare, and business cycles are largely driven by TFP shocks. As such, the response of our economy with a rich distribution of firms broadly resembles a typical equilibrium business cycle model. GDP is more variable than consumption, investment is more variable than output, and both series are strongly procyclical, as is hours worked. Unlike a typical business cycle model, however, this setting has an endogenous number of producers. Movements in the distributions of firms propagates, rather than amplifies recessions. Thus consumption is less correlated with output than it is in a representative firm model. As in U.S. firm-level data, our baseline model's exit rate is countercyclical and its entry rate is procyclical. The percent standard deviation of entry is 0.209 while that of exit is 1.193.

TABLE 6. Baseline Business Cycles with TFP Shocks Only

$x =$	GDP	C	I	N	K	r
$mean(x)$	0.488	0.384	0.104	0.334	1.149	0.042
σ_x/σ_{GDP}	(2.335)	0.440	3.416	0.651	0.576	0.381
$corr(x, GDP)$	1.000	0.875	0.968	0.946	0.090	0.681

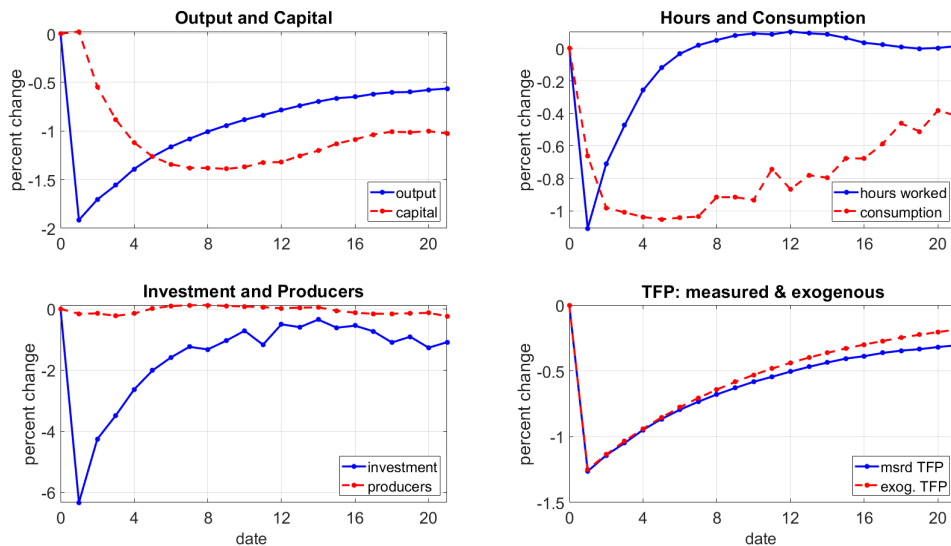
Results of 10,000 period simulation; $corr(exit, Y) = -0.504$; $corr(entry, Y) = 0.227$.

Table 6 examines a version of our baseline model with only aggregate productivity shocks. Comparing this to Table 5, we see that the presence of credit shocks marginally raises the cyclical volatility of GDP a bit and reduces its correlations with consumption, employment and investment somewhat. Entry continues to be weakly procyclical, as in the data, whereas exit is strongly countercyclical. The standard deviation of exit and entry rates are 0.401 and 0.125, respectively. Both are less variable than in the model with credit shocks. Given such shocks are rare events, this suggests that we imply significant changes in the measure of producers. As entrants amplify the misallocation of capital,

this implies that our model drives changes in the endogenous component of aggregate total factor productivity. Aside from this, the high variability exit in our model arises from the cyclicality of default. Overall, business cycle moments across the two tables are similar. We will confirm using impulse responses below that this apparent invariance to financial shocks is entirely because such shocks occur in only 7 percent of all dates and not because fluctuations driven by such shocks are near-isomorphic to those driven by TFP shocks. Those results are foreshadowed to an extent by the larger relative volatilities in employment and investment in Table 5 versus Table 6. Each of these observations applies equally well for the analogues to Tables 5 - 6 from our alternate calibration, including the ordering of volatility and correlation changes, so we omit those tables.

Figure 1 shows our baseline model's impulse responses following a persistent 1.75 percent shock to the exogenous component of total factor productivity. In most respects, the economy's response is similar to that of a representative firm equilibrium business cycle model without financial frictions. The largest response is in investment, and the GDP response exceeds that of consumption. Hours worked, consumption and investment are procyclical. Responses in output, hours and investment are essentially monotone, and consumption exhibits the customary hump. The lower right panel of the figure shows a small gap between the exogenous change in TFP and measured TFP, which widens slightly over time. This happens due to a small, gradual fall in the number of producers (not shown here) and, in turn, causes a slight flattening in the GDP recovery. Responses to the TFP shock in the alternate version of our model (not shown) share these features.

FIGURE 2. TFP-Shock Recession in the Baseline Model



With the persistent decline in productivity, the risk-free real interest rate falls, implying a rise in the risk-free discount rate on loans. However, many firms cannot borrow at that rate, or cannot borrow what they wish. While not shown here, our analysis in section 3.4 implies that the negative TFP shock yields a general worsening of firms' credit situations. Lower productivity raises default probabilities; that reduces firms' ability to borrow at the rates they otherwise would, raising the fraction of type 2 constrained firms and further rationing such firms' investment activities. With this said, the 2 percent gradual decline in the overall level of debt in the economy this generates is similar to that from a model with exogenous collateral constraints (Khan and Thomas (2013)).

A new feature of our environment is its allowance for endogenous changes in the number of producers, and thus a second channel affecting capital allocation. We noted above that the fall in aggregate productivity worsens credit conditions for firms. Given worsened loan rate schedules in light of reduced productivity, alongside lower productivity on its own, firm values conditional on operating fall, driving increased exit. At the impact of the shock, the exit rate rises roughly 0.06 percentage points, reducing the number of producers by 0.2 percent. Thereafter, exit rates continue rising for several periods as firms vulnerable to default continue experiencing a slight worsening in their borrowing terms that renders them more so. Because the same forces prompting these changes also reduce the number of entering firms relative to normal, the overall measure of producers steadily declines over about 7 periods and ultimately reaches 0.7 percent below its starting value. Given decreasing returns to scale at the firm ($\alpha + \nu = 0.865$), these changes deliver a separate source of misallocation beyond that associated with inefficient capital choices among financially constrained continuing firms. That extra source of propagation, changes in the measure of producers, drives the small wedge between exogenous and measured TFP seen in Figure 1.

We next explore aggregate responses in our economy to a financial shock effectively reducing firms' cash-on-hand and raising their costs of borrowing. This shock is an unanticipated change in credit conditions from ζ_o to ζ_l and lasts for 4 periods. While ζ is at its low value in dates 1 through 4, all firms experience additional financial costs, $\xi_1(\varepsilon)$, of operation that amount to balance sheet shocks. Because loan rate schedules are influenced by x (section 3.4), this directly increases the credit frictions facing firms. However, recall that the shock has no direct real effect on the economy, as all defaulting firms' capital not recouped by the intermediary is lump-sum returned to households, and the additional costs reducing firms' balance sheets are purely financial.

FIGURE 3. Credit-Shock Recession in the Baseline Model

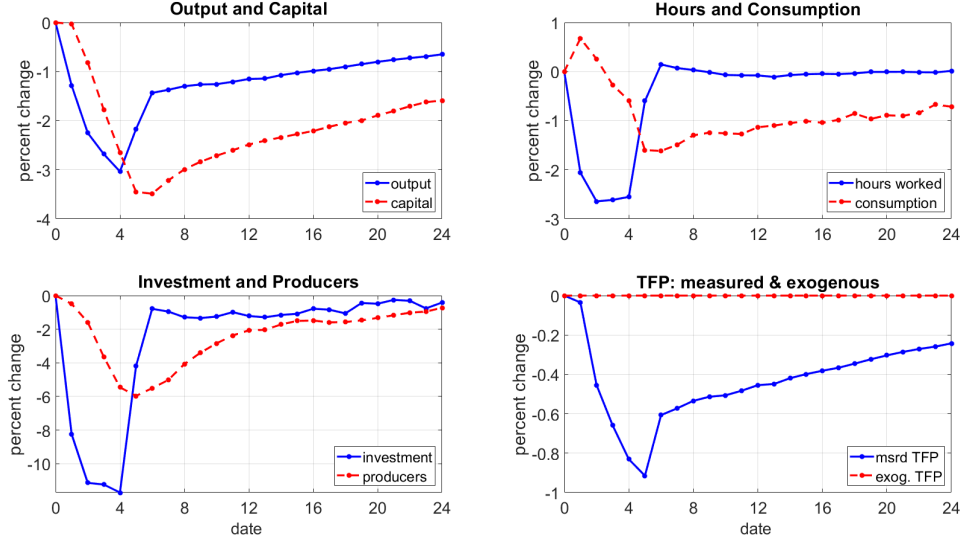


Figure 2 shows the financial shock recession in the baseline case of our model emphasizing the empirical age and size distribution in its calibration. In this case, the financial shock lowers the capital recovery rate in the event of default from 0.37 to 0, and it turns on the financial operating costs $\xi_1(\varepsilon)$ described in section 5, with a 3.5 percent scale factor. Anticipating a rise in default next period due to firms' worsened cash positions, alongside greater loss from each default, the financial intermediary demands higher interest rates for loans involving a nonzero probability of default. This persistently worsens the situations of type 2 financially constrained firms. Moreover, it causes an unusually large number of type 1 firms to become type 2, so their investment decisions are no longer purely a function of productivity. Reductions in the numbers of producers are a second important contributing factor; as noted above, our economy generates procyclical entry and countercyclical exit in response to the financial shock.

The top right and lower left panels in Figure 2 show a small initial rise in employment and investment as households respond to the loss in future income they expect the two forms of increased misallocation will cause. Thereafter, GDP, employment and investment steadily fall over several periods. The driving force behind this is an anticipated decline in endogenous productivity in the lower right panel, implying low returns to investment over several periods. Part of that endogenous TFP declines comes because type 2 firms' capital choices shift further from the frictionless capitals associated with their productivities, and because there are more such firms than usual. However, this increased intensive-margin misallocation among incumbents is amplified by changes in the numbers of firms (production locations) shown in the lower left panel.

The worsening of credit conditions simultaneously raises default thresholds and pushes an unusually large number of firms' cash positions down into regions involving default risk. While not shown here, this drives unusually high exit rates, and unusually low entry rates, for so long as the credit situation does not improve. At the impact of the shock, when capital stocks are already in place, the aggregate exit rate rises about 0.7 percentage points, driving roughly a 3.4 percent fall in the measure of producers in the lower left panel of Figure 2. At the same time, the entry rate falls by almost 2.5 percentage points. Over the next few periods, the entry rate regains some ground, to around 1.75 percentage points below normal, as the number of firms in its denominator shrinks, whereas accumulated damage to type 2 firms' production levels drives a slight further increase in the exit rate. On balance, the number of producers steadily falls over dates 2 - 4, with an ultimate loss of about 11 percent of the economy's producers by the time credit conditions are restored. Given the usual per-period supply of potential entrants and the fact that only a fraction of them ever chose to enter production, it takes many periods for the number of firms to recover from such damage. Because the stock of firms is itself a valuable input for aggregate production, these losses slow the aggregate productivity recovery, and so hold down returns to savings after credit conditions are restored in date 5. That, in turn, delivers sluggish recoveries in investment and employment, slowing the recovery in aggregate capital and, hence, in GDP and consumption.

6.2 The recent recession

Table 7 compares the peak-to-trough changes from our baseline model in response to the TFP and credit shocks described above with observations from the 2007 U.S. recession, and Table 8 presents the same comparison for our alternate model calibration. In each case, we set the size of the TFP shock to match the endogenous TFP fall caused by the credit shock.¹³ We may confine our discussion to the baseline case in Table 7, noting that Table 8 shares very similar features.

¹³The measured TFP drop is marginally higher because we chose the exogenous TFP shock to match the 1.72 percent (1.56 percent) endogenous decline over the credit shock recession without adjusting for the slight rise in misallocation the productivity shock causes by raising default and thus reducing the measure of producers.

TABLE 7. Peak-to-trough Declines: U.S. 2007 Recession and Baseline Model

	trough	GDP	I	N	C	TFP	Debt
data	2009Q2	5.59	18.98	6.03	4.08	2.18	25.94
credit shock*	5	3.03	11.73	2.55	0.59	0.83	39.23
tfp shock	1	1.93	6.31	1.10	0.66	1.27	0.00

*credit shock: 4-pds of $\tau = 0.97$ (versus 1.0) and 1.0%-scaled financial operating costs

The data row in Table 7 reports seasonally adjusted, HP-filtered real quarterly series and measures declines between 2007Q4 and 2009Q2. The one exception to this is the debt entry, where we report the ultimate drop in the stock of real commercial and industrial loans, which came later. The credit shock row reports the declines in our model in response to the credit shock described above. Declines in real series are reported as of the GDP trough date, period 5. In contrast to other real series, the declines in consumption continue for several further periods, as GDP recovers gradually while the rebuilding of the capital stock begins. Consumption ultimately reaches 2.53 percent below average around date 10. As in the data, debt in the model falls more slowly than GDP, and we report the ultimate decline at its trough, which occurs in date 8.

The losses in GDP, investment, employment and debt are disproportionate in the credit shock row relative to the endogenous fall in measured TFP. This unusual aspect of our model's response to a credit shock response resembles that over the 2007 recession. In our model, it appears to stem from the strong disincentives for investment in both physical capital and firms when misallocation not only worsens, but is anticipated to worsen further. The non-monotone path of measured TFP (despite the monotone path of ζ) itself happens as an increasing number of young cohorts are affected by tightened credit conditions, and vulnerable incumbents are forced to exit.

By contrast, our model's GDP, investment and employment responses to the TFP shock are monotone. Thus, the GDP trough occurs at date 1 in the TFP shock row, and we report date 1 declines for all series except debt. Debt falls negligibly at date 1, so here again we report the ultimate drop in that series, which happens 5 periods after the impact of the TFP shock.

The credit and TFP shocks are equivalent in their effects on measured productivity; both model rows capture roughly 80 percent of the observed decline. However, there are large differences in other columns. The credit shock drives roughly 84 percent of the observed reduction in GDP, 67 percent of the observed decline in hours worked, 125 percent of the empirical drop in investment, and about 37 percent of the ultimate decline in debt. By contrast, confronting the same model with

the same-sized exogenous TFP shock delivers only 46 percent of the reduction in GDP seen over the 2007 recession. The fall in investment there is also about half that in the data, there is no significant reduction in debt, and the fall in employment is less than one-fourth the observed decline.

7 Concluding remarks

We have developed a model where the aggregate state includes a distribution of firms over idiosyncratic productivity, capital and debt. Firms are risky borrowers, and equilibrium loan rate schedules reflect each borrower's default risk. We associate default with exit, and find that our model economy's response to a persistent shock to exogenous total factor productivity is similar to a representative firm business cycle model without financial frictions. In contrast, while our economy is consistent with the average aggregate debt-to-asset ratio in the U.S., and only a small fraction of production takes place in firms with investment hindered by financial frictions ordinarily, a credit shock affecting firms' balance sheets and debt recovery rates in the event of default generates a large and lasting recession with some unique features.

Our model's credit shock recession unravels through two sources of capital misallocation that grow over several periods. First, the misallocation of capital across continuing firms gradual worsens as several subsequent cohorts of young firms with little cash but high expected future productivity encounter worsened borrowing terms. Second, there are large endogenous reductions in firm entry and increases in exit following the credit shock that sharply reduce the number of firms over the downturn. Because the number of production locations is itself a valuable input affecting the productivity of the aggregate stock of capital, this second misallocation effect compounds the first in moving the distribution of production further from the efficient one and reducing measured productivity. Furthermore, because this substantial damage to the stock of firms takes time to repair, our model delivers a far more gradual rebuilding of the aggregate capital stock, and thus more gradual recoveries in employment and investment than would otherwise occur.

Despite marked differences in response to real versus financial shocks, our setting delivers business cycle moments similar to a snapshot of postwar U.S. business cycles, because credit shocks in our model, as in the data, are infrequent. While aggregate responses following a productivity shock are unconstrained to the details of micro-level calibration, we have seen that the average age-size distribution in the model affects its responses to a credit shock in important ways. The explanation for this rather unique aspect of our model lies in the fact that, whereas an aggregate TFP shock's consequences are for the most part level across firms, a financial shock has disparately large effects

on young, small firms most reliant on external finance and most vulnerable to default and exit.

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